

Echoes of Heron

Drayan Mathematical Series

Problem

Let $(x, y, z) \in \mathbb{R}^3$ satisfy

$$-x^2 + xy + xz = 4,$$

$$-y^2 + xy + yz = -20,$$

$$-z^2 + xz + yz = -8.$$

Determine all real triples (x, y, z) satisfying the system.